Litmus: video script

About 616 words · roughly 3:58 at a normal pace · one section per slide

1 · COVER

LLMs are increasingly used to explain what molecular models have learned. They write confident chemistry, like “primary amines make molecules more soluble.” But how often are those claims actually true? To find out, I built Litmus: a pipeline that turns every claim into a test, and that anyone can run.

2 · SHORT ANSWER

The short answer: most claims pass a simple check on new data. Far fewer survive once you account for the obvious explanation. And when the LLM was asked to fix a weak rule, it reversed the rule and invented a reason for it.

3 · WORKFLOW

Here’s what the pipeline does, in four steps. First, it trains a model to predict a molecular property, and then generates counterfactuals, which are slightly changed versions of a molecule that can alter the model’s prediction. They reveal which parts of a molecule the model is actually paying attention to. Second, an LLM reads those counterfactuals and writes rules, each one a claim a computer can check. Third, every rule is tested on held-out molecules the LLM never saw. And fourth, each claim is checked against published papers, found with Bond Valency. So what happens when we run it on real data? To find out, I tested it on the datasets provided for this hackathon.

4 · RESULTS

These are the results of my test runs on the provided datasets: ESOL, logD and FreeSolv from the hackathon’s list, plus a larger solubility set, AqSolDB. Each rule gets two tests. The first is direct: do the molecules that match the rule really differ? The second is harder. Solubility and logD are largely explained by lipophilicity, and many rules just restate it, so we compare each molecule only with others of similar lipophilicity and ask whether the rule still holds.

In total, the LLM proposed 30 rules across the four datasets.

Orange is the direct test, teal the harder one. Most rules pass the first, far fewer the second. LogD is the exception, where the LLM correctly spotted that charged acids are less lipophilic than predicted. And a few rules, in red, are flatly contradicted.

5 · REVISION

Here's one rule from start to finish. The LLM said primary amines make molecules more soluble. On the validation data that was right overall, but it didn't hold at matched lipophilicity, so the LLM got one chance to revise it. The right fix would have been "no extra effect." Instead, it reversed the claim, saying amines make molecules *less* soluble because of crystal lattice energy. On the test data that's clearly wrong: off by a full log unit, at p of four times ten to the minus twelve, across 175 molecules. This happened three times on two datasets.

6 · LITERATURE

So did the published papers catch that mistake? This time, yes: a 1999 study found that hydrogen bonding always *increases* solubility, which matches the data. But across all four datasets, papers and data sometimes disagreed, and in a revealing way. Take "molecules with no hydrogen-bond donors have higher logD." The data say that's true, but the LLM's explanation for it is wrong, so the literature check rejects it. Or "hydrazones are less soluble": the data say it's false, but the explanation sounds like textbook chemistry, so the literature check accepts it. Papers judge the reasoning; data judge the result. You need both.

7 · CLOSE

So here are the key limitations. Due to time constraints, we only ran one run per dataset, generated about eight rules each, and had varied LLM outputs between runs. But the testing framework is the contribution: any molecular dataset plus one config file gets the same checks. So if you use an LLM to explain a chemistry model, test its claims beyond the obvious, and don't trust it to fix its own mistakes.